RAG Learning

4.3 Transformations

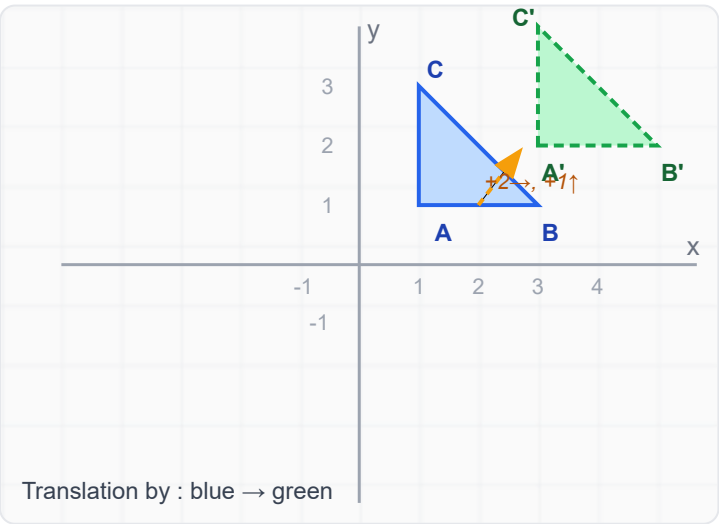
OCR GCSE Mathematics J560 — Component 4 (Geometry and Measures) — Papers 1, 2, 3

4.3.1 Translation

Translation moves every point of a shape the same distance in the same direction. The shape does not rotate or reflect — size and orientation are preserved. Translations are described using a **column vector** $\begin{pmatrix} x \\ y \end{pmatrix}$ where x is the horizontal shift (positive = right) and y is the vertical shift (positive = up).

Under a translation, the image is **congruent** to the object: same shape, same size, same orientation.

Column vector notation:
$\begin{pmatrix} a \\ b \end{pmatrix}$ means: move a units right (negative = left), b units up (negative = down).



Worked Example: Triangle PQR has vertices P(1, 2), Q(4, 2), R(1, 5). It is translated by $\begin{pmatrix} 3 \\ -4 \end{pmatrix}$. Find the coordinates of the image.

Add the column vector to each vertex:
P' = (1+3, 2−4) = **(4, −2)**
Q' = (4+3, 2−4) = **(7, −2)**
R' = (1+3, 5−4) = **(4, 1)**

Exam Tip: When describing a translation, you **must** state the column vector — saying "3 right and 4 down" scores no marks by itself unless the vector is given. Always write it as $\begin{pmatrix} 3 \\ -4 \end{pmatrix}$.

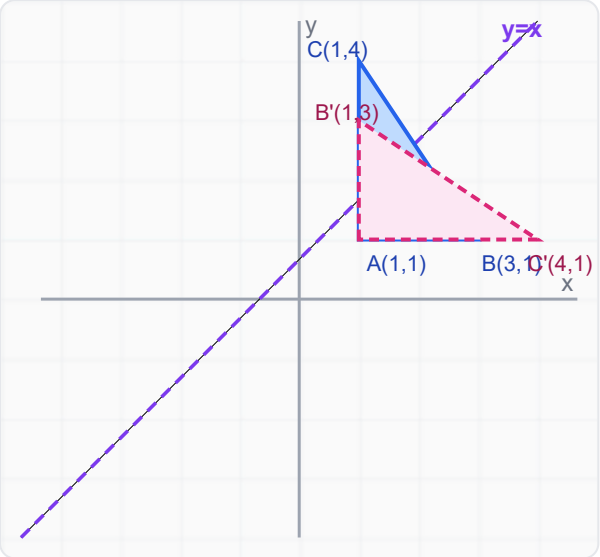
4.3.2 Reflection

A **reflection** maps each point to its mirror image in a given line (the **mirror line**). The perpendicular distance from any point to the mirror line equals the perpendicular distance from its image to the mirror line. The image is congruent but the orientation is reversed (laterally inverted).

Common Mirror Lines

Mirror Line	Rule: $(x, y) \rightarrow$	Example
x-axis ($y = 0$)	$(x, -y)$	$(3, 2) \rightarrow (3, -2)$
y-axis ($x = 0$)	$(-x, y)$	$(3, 2) \rightarrow (-3, 2)$
$y = x$	(y, x)	$(3, 2) \rightarrow (2, 3)$
$y = -x$	$(-y, -x)$	$(3, 2) \rightarrow (-2, -3)$

$y = k$ (horizontal)	$(x, 2k - y)$	Reflect in $y = 3$: $(x, 6 - y)$
$x = k$ (vertical)	$(2k - x, y)$	Reflect in $x = 2$: $(4 - x, y)$



Reflection in $y = x$: blue object → pink image. Coordinates swap: $(x, y) \to (y, x)$.

Worked Example: Triangle with vertices A(2, 1), B(5, 1), C(5, 4) is reflected in the line $y = -x$. Find the image vertices.

Rule for $y = -x$: $(x, y) \to (-y, -x)$
A(2,1) → A'(−1, −2)
B(5,1) → B'(−1, −5)
C(5,4) → C'(−4, −5)

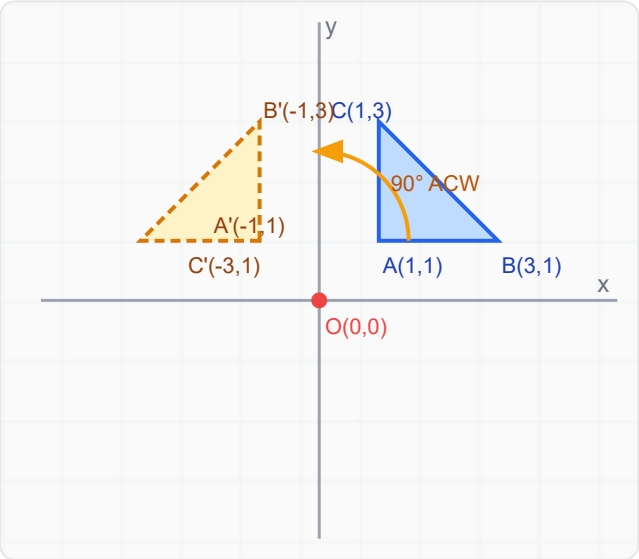
Exam Tip: When describing a reflection, you **must** state the equation of the mirror line (e.g. "reflection in the line $y = x$"). Saying "reflected" alone scores 0. Check the mirror line passes through the midpoint of each object-image pair.

4.3.3 Rotation

A **rotation** turns a shape through an angle about a fixed point called the **centre of rotation**. To fully describe a rotation, state: (1) the centre, (2) the angle, (3) the direction (clockwise or anticlockwise). A 180° rotation requires no direction.

Standard Rotation Rules (centre at origin)

Rotation	$(x, y) \rightarrow$
90° anticlockwise	$(-y, x)$
90° clockwise	$(y, -x)$
180°	$(-x, -y)$
270° anticlockwise (= 90° clockwise)	$(y, -x)$



Worked Example: Triangle with vertices P(2,0), Q(5,0), R(2,3) is rotated 90° clockwise about the origin. Find the image.

Rule for 90° clockwise: $(x, y) \rightarrow (y, -x)$
P(2,0) → P'(0, −2)

Q(5,0) → Q'(0, −5)
R(2,3) → R'(3, −2)

Finding the Centre of Rotation

Draw the perpendicular bisectors of the lines joining each object vertex to its image vertex. These bisectors intersect at the centre of rotation.

Exam Tip: A 270° anticlockwise rotation is the same as a 90° clockwise rotation. Always state direction unless the angle is 180°. The centre does not have to be the origin — check by measuring distances.

4.3.4 Enlargement

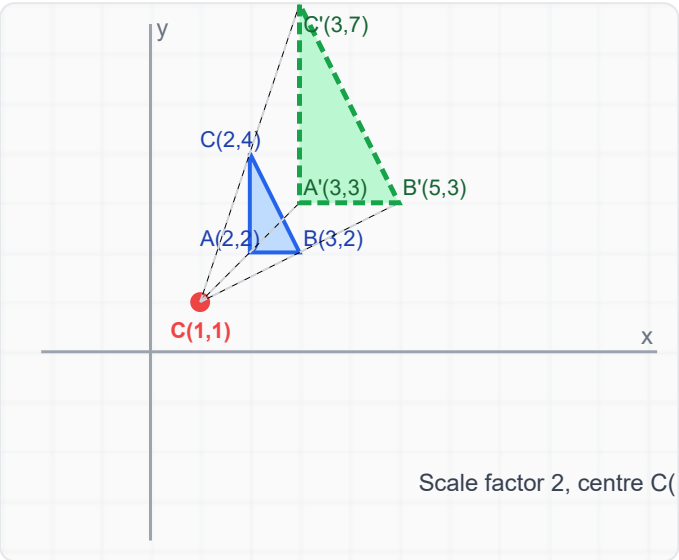
An **enlargement** changes the size of a shape. It is defined by a **centre of enlargement** and a **scale factor** k. Each point is mapped so that its distance from the centre is multiplied by k. The image is **similar** (not congruent unless $|k| = 1$) to the object.

Types of Scale Factor

Scale Factor k	Effect
$k > 1$	Enlargement (image larger, same side of centre)
$0 < k < 1$	Reduction (image smaller, same side of centre)
$k = 1$	No change (image = object)
$k < 0$ H	Enlargement on opposite side of centre; image inverted

To enlarge from centre C with scale factor k:
Vector from C to image point = $k\times$ (vector from C to object point)

$$\overrightarrow{CA'} = k \cdot \overrightarrow{CA}$$



Worked Example: Positive SF: Triangle A(1,1), B(3,1), C(1,4) enlarged by scale factor 2, centre (0,0).
Multiply each coordinate by 2:
A' = (2, 2), B' = (6, 2), C' = (2, 8)

Fractional SF: Same triangle, scale factor $\frac{1}{2}$, centre (0,0).
A' = (0.5, 0.5), B' = (1.5, 0.5), C' = (0.5, 2)

Worked Example: H **Negative SF:** A(2, 1), B(4, 1), C(2, 3), enlarged scale factor -2, centre (1, 1).
Vector C to A: $(2-1, 1-1) = (1, 0)$; multiply by -2: $(-2, 0)$; A' = $(1-2, 1+0) = (-1, 1)$
Vector C to B: $(3, 0)$; multiply by -2: $(-6, 0)$; B' = $(1-6, 1) = (-5, 1)$
Vector C to C: $(1, 2)$; multiply by -2: $(-2, -4)$; C' = $(1-2, 1-4) = (-1, -3)$
Image is on the opposite side of the centre and inverted.

Exam Tip: To find the scale factor from a diagram: measure the length of a side on the image, divide by the corresponding length on the object. To find the centre: draw straight lines

through corresponding vertices — they all meet at the centre.

4.3.5 Describing Transformations

When asked to **fully describe** a single transformation, you must give all required information:

Transformation	Required information
Translation	The word "translation" + column vector $\begin{pmatrix} a \\ b \end{pmatrix}$
Reflection	The word "reflection" + equation of the mirror line
Rotation	The word "rotation" + centre + angle + direction (unless 180°)
Enlargement	The word "enlargement" + scale factor + centre of enlargement

Exam Tip: Always use the correct transformation name. Partial answers (e.g. "it has been flipped") score very few marks. Check: are angles preserved? (Yes for all four.) Is size preserved? (Only for translation, reflection, rotation.) Is orientation preserved? (No for reflection; no for negative SF enlargement.)

How to Identify the Transformation

- **Same size, same orientation** → Translation
- **Same size, reversed orientation (mirror image)** → Reflection
- **Same size, rotated** → Rotation
- **Different size, same angles** → Enlargement

4.3.6 Combined Transformations

A **combined transformation** applies two or more transformations in sequence. The order matters: transformations are generally **not commutative** (doing A then B is usually different from doing B then A).

Worked Example: Triangle T has vertices (1,0), (3,0), (1,2). Apply: (1) Reflection in the y-axis, then (2) Translation $\begin{pmatrix} 0 \\ -3 \end{pmatrix}$.

Step 1 — Reflect in y-axis: $(x, y) \to (-x, y)$
(1,0)→(−1,0), (3,0)→(−3,0), (1,2)→(−1,2)

Step 2 — Translate by $\begin{pmatrix} 0 \\ -3 \end{pmatrix}$:
(−1,0)→(−1,−3), (−3,0)→(−3,−3), (−1,2)→(−1,−1)

Final image: **(−1,−3), (−3,−3), (−1,−1)**

Worked Example: Show that the combined transformation above is NOT the same if the order is reversed.

Step 1 — Translate first: (1,0)→(1,−3), (3,0)→(3,−3), (1,2)→(1,−1)
Step 2 — Reflect in y-axis: (1,−3)→(−1,−3), (3,−3)→(−3,−3), (1,−1)→(−1,−1)

Final image: **(−1,−3), (−3,−3), (−1,−1)** — same result in this case.
(This particular pair of transformations happens to commute. Reflections and rotations about different centres do not in general.)

Exam Tip: In a two-transformation question, always label the intermediate image (e.g. T') and the final image (e.g. T''). Apply transformations one at a time — do not try to combine the algebra. Show all intermediate coordinates.

Key Properties Summary

Property	Translation	Reflection	Rotation	Enlargement
Congruent?	Yes	Yes	Yes	Only if $k = \pm 1$
Similar?	Yes	Yes	Yes	Yes
Orientation preserved?	Yes	No	Yes	Yes (unless $k < 0$)
Angles preserved?	Yes	Yes	Yes	Yes

© RAG Learning · Not for redistribution